

Fig. 2: Circuit diagram of the simple audio buffer for guitar and line driver with low quiescent current

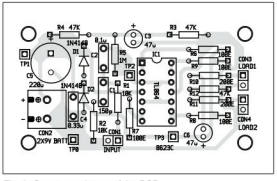


Fig. 3: Component layout of the PCB

much higher impedance, for example, 10-kilo-ohm or more. If two loads of 1-kilo-ohm or higher are to be driven, then these can be connected simultaneously to CON3 and CON4.

The peak-to-peak amplitude of the signal from the pickups and the minimal driven load should be taken into consideration when power supply is selected.

Although the circuit will work with single 9V battery (6F22), it is preferable to use at least two 9V batteries producing 18V. But 12V and 24V rechargeable car batteries can also be used.

The circuit does not require any adjustment and short circuit at the outputs (CON3 and CON4) will not damage the buffer. It can drive 600-ohm line with a length of 10m or more.

Construction and testing

An actual-size, single-side PCB layout of the audio buffer and line driver with low quiescent current circuit is shown in Fig. 2 and its component layout in Fig. 3.

After assembling the components on the PCB, connect the guitar input to CON1 and connect CON3 to the next equipment, such as the sound amplifier. Connect 18V DC supply at CON2; now you are ready to go with the guitar.

For troubleshooting, check various voltages at test points given in the table.

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